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## Executive Summary

The Community Data System (CDS) is a digital public good designed to help local communities collect, safeguard, and act on data that matters to their neighbourhoods — from infrastructure faults and public service gaps to environmental observations. Residents can submit data through a simple web portal, community researchers can run privacy-respecting analysis on anonymized datasets, and local administrators can turn insights into action. The system directly supports United Nations Sustainable Development Goal 11 (Sustainable Cities and Communities) by giving under-resourced communities the same data infrastructure that better-funded municipalities already enjoy, enabling more inclusive, participatory, and evidence-based local governance.

This report documents the complete software design of the Community Data System, building on the group's earlier Digital Public Good concept. It covers the Software Development Life Cycle (SDLC) approach, system architecture, prototype design, GitHub-based version control workflow, project plan, licensing, and SDG alignment, in line with the PROG 102 Final Project brief.

## 1. Software Development Life Cycle (SDLC)

The Community Data System is being developed using the Agile (Scrum) SDLC model. Agile was chosen over a traditional Waterfall approach because the project has a fixed academic deadline (Week 11) but an evolving understanding of community needs, data-privacy requirements, and UI feedback. Agile allows the group to work in short, reviewable cycles (sprints), incorporate feedback from lecturers and peer testers between sprints, and still deliver a working prototype even if later features slip.

### 1.1 Why Agile (Scrum) Fits This Project

- Requirements around data anonymization and permissions were expected to change as the group researched privacy best practice — Agile accommodates this through iterative refinement.
- A working, demonstrable prototype was required by Week 11 regardless of

scope changes; Scrum's sprint reviews keep a shippable increment available at all times.

- The four-person group could split into parallel workstreams (architecture/backend design, UI/UX, documentation, project management) that Scrum's daily stand-ups and sprint boards coordinate naturally.

## 1.2 SDLC Phases Applied to the Community Data System

### *Phase 1: Requirements Gathering & Planning (Week 4)*

The group identified the core problem: communities in Sierra Leone often lack a simple, trustworthy channel to report local issues and have that data actually inform decisions. User roles (Community Member, Data Researcher, System Admin) and core use cases (Submit Data, View Local Insights, Conduct Research Analysis, Manage Data Permissions, Export Anonymized Reports) were captured in the use case diagram (Figure 3).

### *Phase 2: System Design (Weeks 4–5)*

The group designed the three-tier system architecture (Web Portal, Data API, Database), the data-submission sequence flow (Figure 1), and the research-analysis workflow including mandatory anonymization checks (Figure 2). Database schema, API endpoints, and session-validation logic were defined during this phase.

### *Phase 3: Implementation / Prototype Development (Weeks 5–8)*

Key screens were built as an interactive Figma prototype covering data submission, dashboards, and admin permission management, allowing the design to be user-tested before any backend code is finalized.

### *Phase 4: Testing (Weeks 8–9)*

Prototype walkthroughs were run with peers acting as community members and researchers to check that the submission flow, anonymization step, and insight dashboards were understandable without instructions. Findings were logged as GitHub issues and fed back into the design.

### *Phase 5: Deployment & Documentation (Weeks 9–11)*

The final prototype, diagrams, and this report are packaged and pushed to the group's GitHub repository, alongside the project plan (Gantt chart and network diagram) and licensing documentation, ready for submission and presentation.

### *Phase 6: Maintenance (Beyond Week 11, Conceptual)*

Although outside the assignment's timeline, the design anticipates ongoing maintenance: periodic review of anonymization rules as regulations evolve, community feedback channels for reporting bugs, and versioned releases managed through the same GitHub repository.

## **2. System Architecture Design**

The Community Data System follows a three-tier client–server architecture, separating the presentation layer, application logic, and data storage. This separation keeps the system easy to secure (only the Data API touches the database), easy to scale (the Web Portal and Data API can be deployed independently), and easy to maintain (each tier can be updated without rewriting the others).

### **2.1 Architecture Layers**

- **Presentation Layer – Web Portal:** the interface community members, researchers, and admins use in a browser to submit data, view insights, and manage permissions.
- **Application Layer – Data API:** validates sessions, enforces anonymization and permission rules, and mediates every request between the Web Portal and the Database.
- **Data Layer – Database:** stores submitted data, anonymized datasets, user accounts, and permission records.

### **2.2 Component Interaction – Submit Data Flow**

Figure 1 shows the sequence of a Community Member submitting data. The Web Portal first validates the member's session with the Data API before any payload is accepted, preventing unauthenticated submissions. Once the session is confirmed valid, the portal forwards the data payload to the API, which stores it in the Database and returns a stored confirmation. Only after this round-trip succeeds does the Web Portal show the member a success message — ensuring the user never sees a false confirmation.

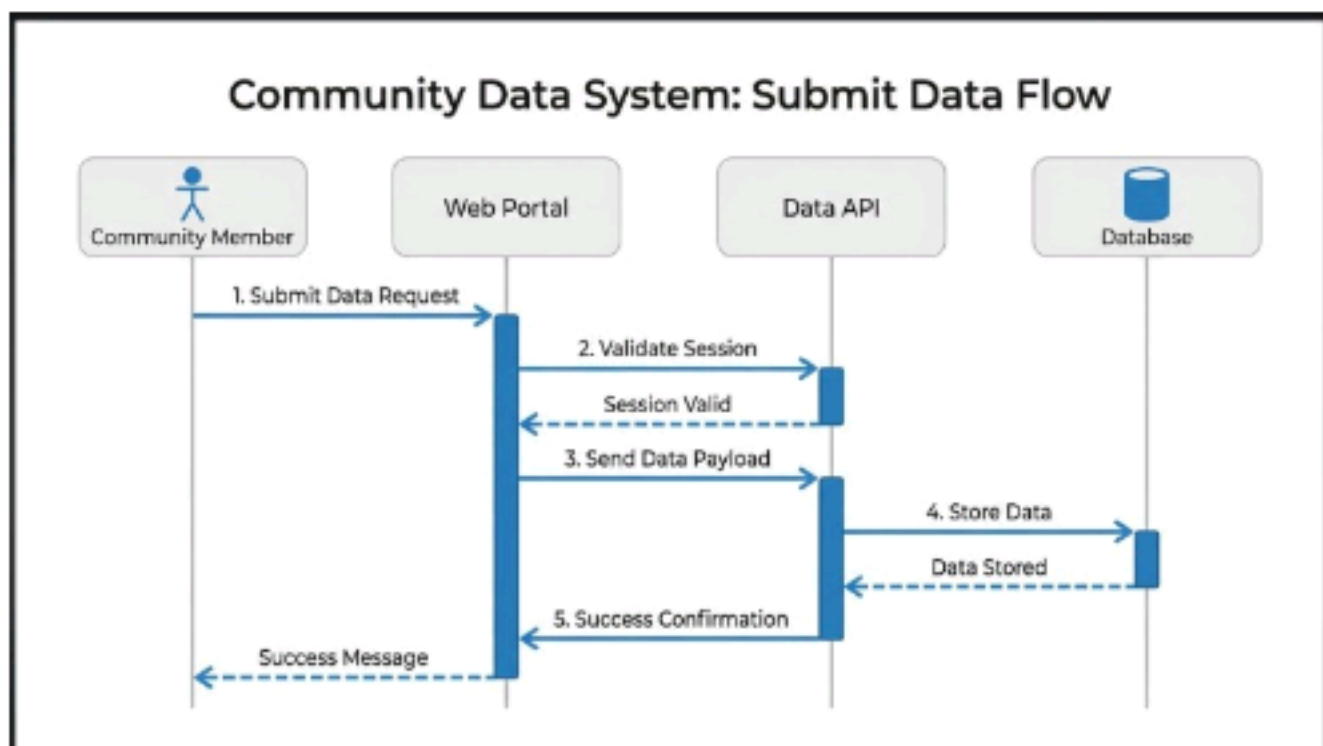


Figure 1: Sequence diagram – Community Data System Submit Data flow

## 2.3 Component Interaction – Use Cases

Figure 3 maps how each actor interacts with the system. Community Members can submit data and view local insights derived from aggregated, anonymized submissions. Data Researchers conduct deeper research analysis and export anonymized reports. The System Admin oversees data permissions and supports research analysis and exports, acting as the accountable party for privacy compliance.

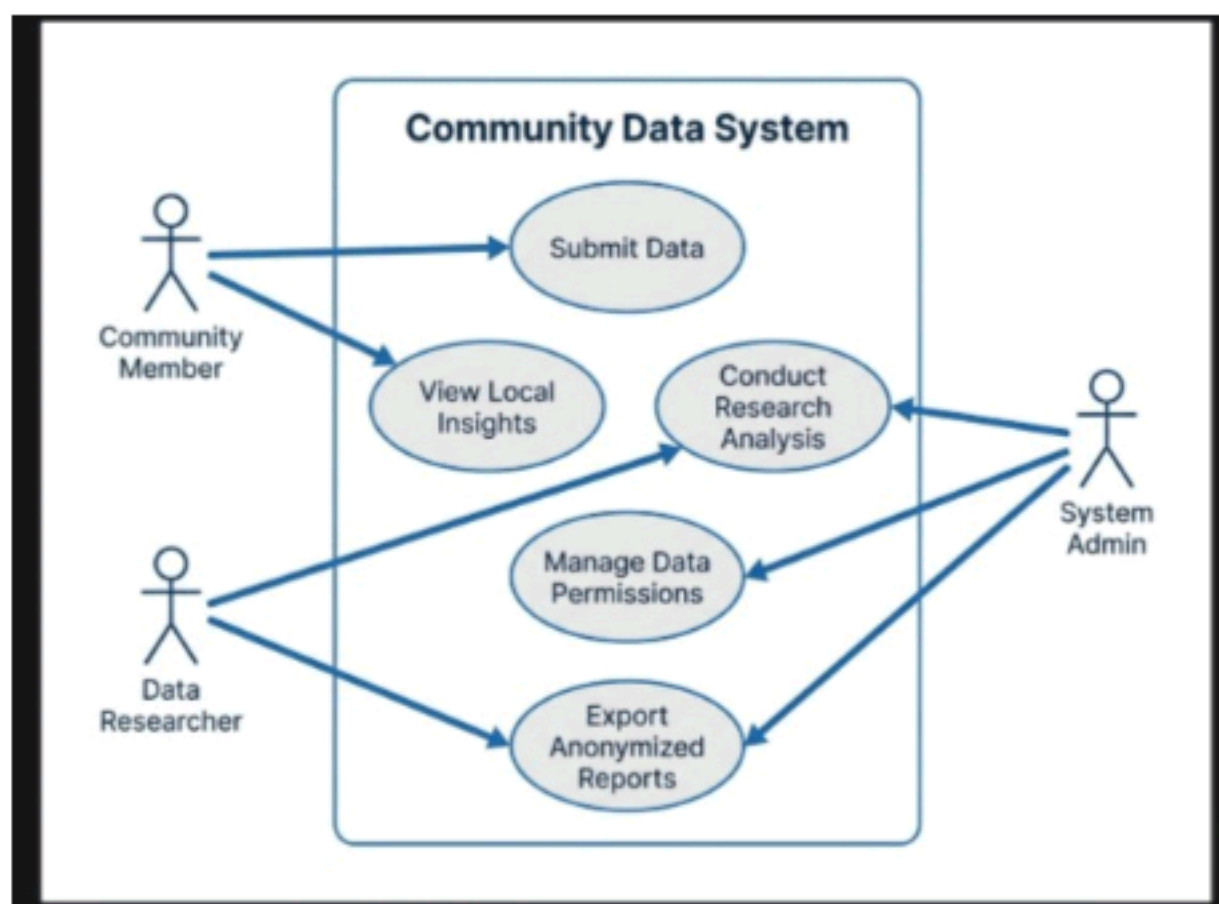


Figure 3: Use case diagram – Community Data System actors and use cases

## 2.4 Design Rationale

Separating “View Local Insights” (available to any community member) from “Conduct Research Analysis” (restricted to researchers and admins) reflects a privacy-by-design principle: everyday users only ever see aggregated, non-identifiable insights, while deeper analysis of anonymized datasets is limited to accountable roles. This directly supports the anonymization safeguard shown in Figure 2.

## 3. Prototype Development

The prototype was designed in Figma as a clickable high-fidelity mockup covering the full journey of each user role, from data submission through to exported insight reports. The prototype also encodes the mandatory anonymization safeguard shown below.

### 3.1 Research Analysis Workflow

Before any dataset can be used for research analysis, the system checks whether it has already been anonymized. If not, anonymization is applied automatically before analysis proceeds – researchers cannot bypass this step. Figure 2 illustrates this decision logic, which was translated directly into the corresponding prototype screens (dataset selection → anonymization notice → analysis → insights → export).

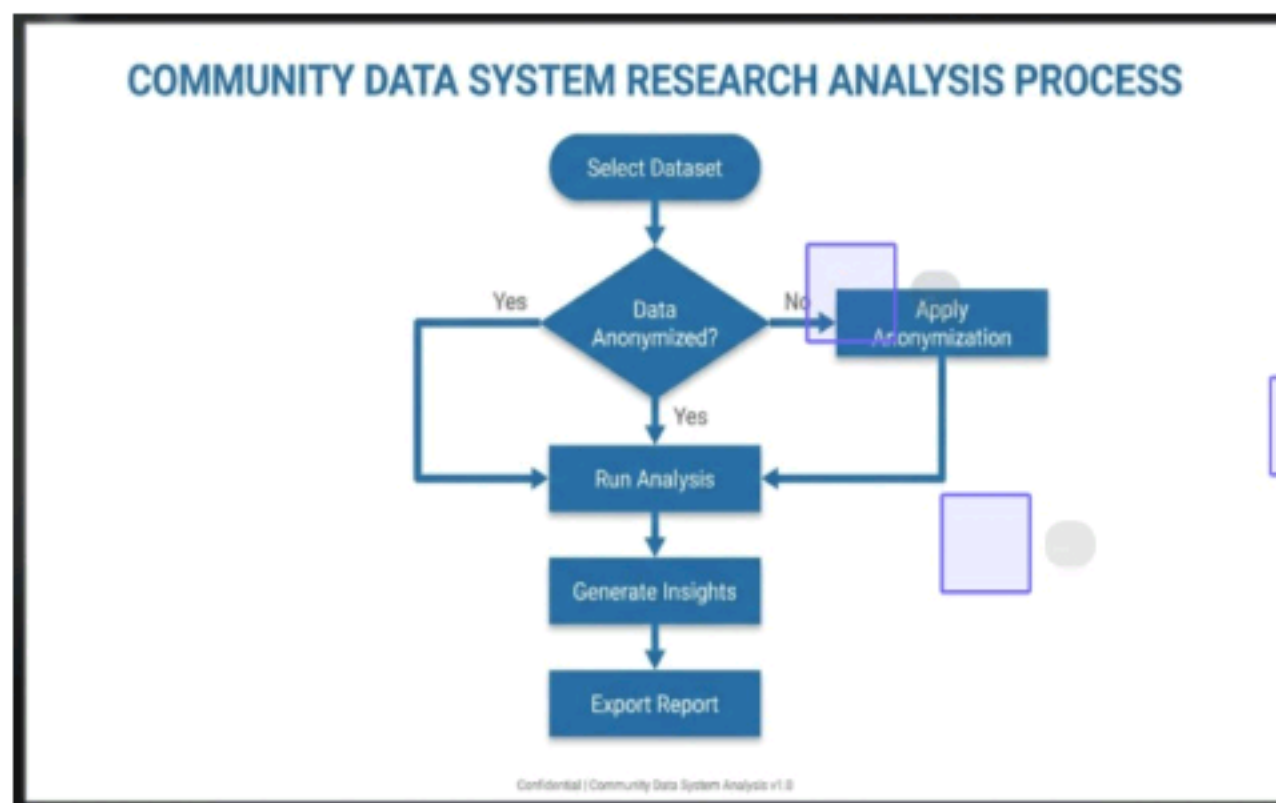


Figure 2: Flowchart – Community Data System research analysis process

### 3.2 Key Screens (Figma UI/UX Design)

At least ten key screens were designed to cover every use case identified in Figure 3:

- 1. Landing / Welcome screen with role-based sign-in
- 2. Community Member registration and login
- 3. Submit Data form (category, description, location, optional photo)
- 4. Submission confirmation screen
- 5. Local Insights dashboard (aggregated, anonymized view for community members)
- 6. Data Researcher dashboard with dataset selection
- 7. Anonymization status / consent notice screen
- 8. Research analysis workspace (filters, charts)
- 9. Export Anonymized Report screen (PDF/CSV export options)
- 10. System Admin – Manage Data Permissions screen
- 11. System Admin – User & role management screen

Note for the group: insert the Figma prototype share link here once finalized, e.g. “Interactive prototype: [Figma link]”, and add 2–3 annotated screenshots per screen category if the module requires visual evidence beyond the link.

### 3.3 UI/UX Design Principles Applied

- Consistency: the same blue/grey colour palette and component style used across all screens (matching the diagrams in this report).
- Clarity over density: each screen focuses on one task (submit, view, analyze, or manage) rather than combining functions.
- Progressive disclosure: raw data and anonymization controls are hidden from community members and only surfaced to researchers/admins who need them.
- Accessibility: high-contrast text, large tap targets, and simple language suited to varying levels of digital literacy.

## 4. Git & GitHub Integration

The group created a shared GitHub repository named community-data-system to host all project artefacts and to demonstrate version control practice, as required by the brief.

### 4.1 Repository Structure

- /docs – this report, UML diagrams (Figures 1–3), and the project plan images
- /prototype – exported Figma screens and the prototype share link
- /src – placeholder / early implementation code for the Web Portal and Data API
- /planning – Gantt chart, network diagram, and sprint notes
- README.md – project overview, SDG alignment, setup instructions, and team member list
- LICENSE – the project's chosen open-source license (see Section 6.1)

### 4.2 Collaboration Workflow

Each group member works on a feature branch (e.g. feature/usecase-diagram, feature/prototype-screens) and opens a Pull Request into main for the other members to review before merging. Commit messages follow a short, descriptive convention (e.g. “Add sequence diagram for submit data flow”). Issues are used to track outstanding tasks and feedback gathered from prototype testing, keeping a visible history of decisions for the examiner to review.

## 5. Project Plan

The project plan translates the SDLC phases into a concrete week-by-week schedule against the brief's Week 4 (issue) to Week 11 (submission) window, and shows how tasks depend on one another.

### 5.1 Gantt Chart

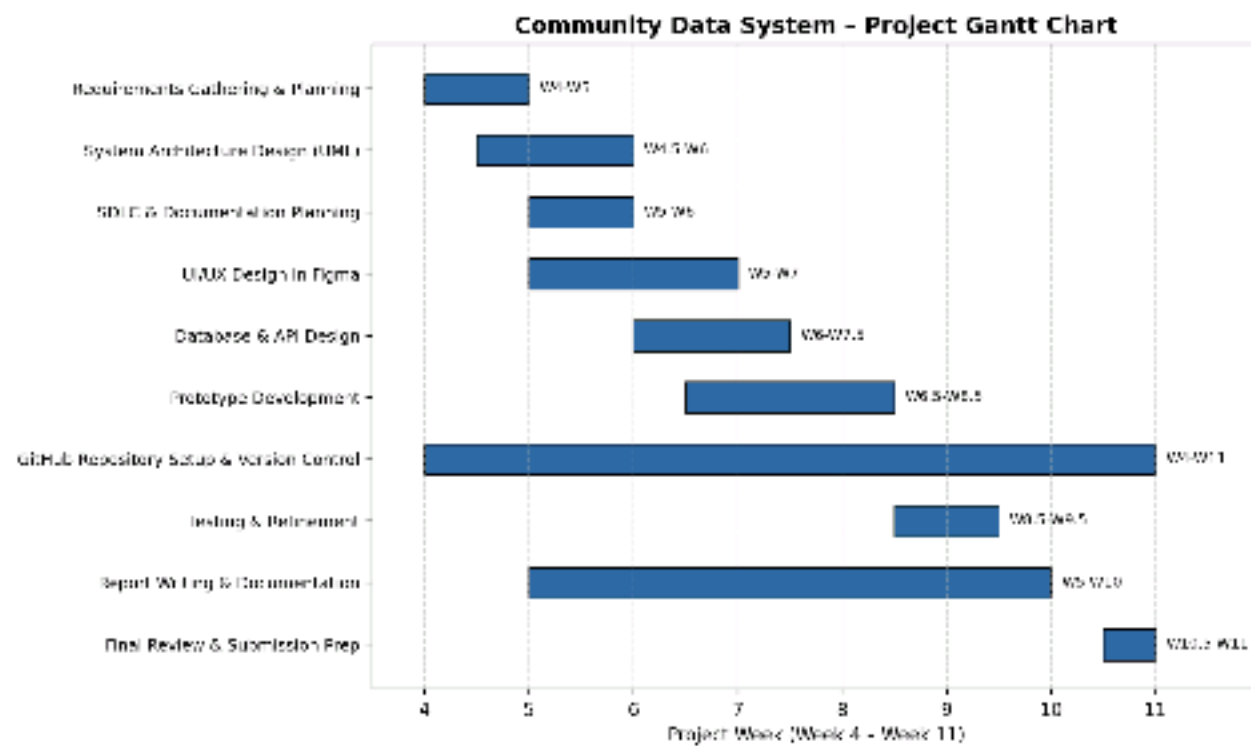


Figure 4: Project Gantt chart (Week 4 – Week 11)

Architecture design and Figma UI/UX work run in parallel once requirements are settled, since the visual design does not have to wait on backend/API decisions. GitHub set-up and report writing run continuously across the whole timeline rather than as single blocks, reflecting real collaborative practice.

## 5.2 Network Diagram

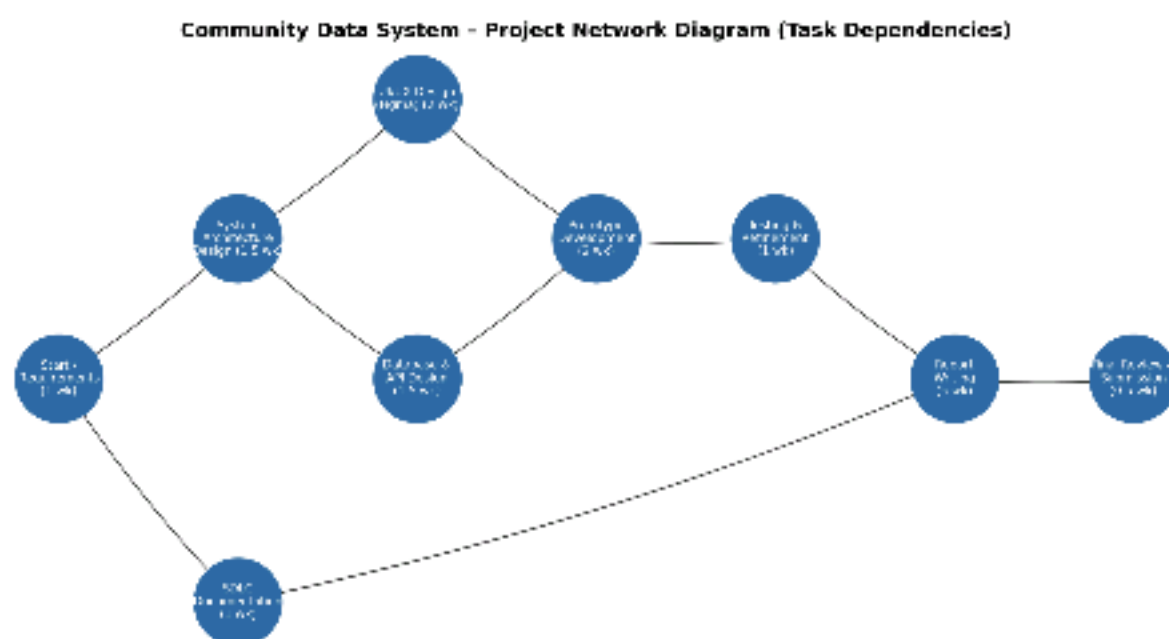


Figure 5: Project network diagram showing task dependencies

The network diagram highlights the critical path: Requirements → Architecture

Design → (UI/UX Design and Database/API Design in parallel) → Prototype Development → Testing → Report Writing → Final Review. Any delay along this path directly delays submission, so the group prioritized architecture sign-off early to protect the downstream prototype and testing work.

## **6. License, Design Quality & SDG Relevance**

### **6.1 License**

The Community Data System is released under the MIT License. MIT was chosen because it is permissive and simple: it allows other communities, NGOs, or student groups to reuse, adapt, and redeploy the system for their own local context — fitting for a “digital public good” that should be freely extendable — while still requiring attribution back to the original authors. The LICENSE file is included in the GitHub repository root. Any external assets used (icons, fonts, or diagram templates) are credited in a CREDITS section of the README with links to their original sources and licenses.

### **6.2 Design Quality (UI/UX)**

Design quality was assessed against three criteria: visual consistency (shared colour palette and component styling across all Figma screens and diagrams), ease of navigation (no screen requires more than two taps/clicks to reach from the home screen for its role), and user experience (the anonymization safeguard and session validation happen automatically, so users are protected by default rather than needing to remember privacy steps themselves).

### **6.3 SDG Relevance – SDG 11: Sustainable Cities and Communities**

The Community Data System supports SDG 11 by strengthening participatory, data-informed local governance, particularly Target 11.3, which calls for enhanced inclusive and participatory human settlement planning and management. By giving

residents a direct channel to report local conditions, and giving administrators anonymized, aggregated evidence to act on, the system helps under-resourced communities in Sierra Leone build the same kind of civic data infrastructure that better-funded cities already have — without requiring residents to hand over identifiable personal data.

- Real-world impact: local issues (e.g. broken infrastructure, service gaps) become visible to decision-makers faster and with community-sourced evidence.
- Problem-solving value: the anonymization-first design lets communities benefit from data analysis while limiting privacy risk, addressing a common barrier to civic data projects in low-resource settings.

 Community Data



Empowering

 Community Data



## Welcome back, Alex

Here is a snapshot of your community contributions today.

MY CONTRIBUTIONS

1,248

+12% from last month

COMMUNITY PARTICIPATION

85.4%

Top 5% active members

LATEST INSIGHTS

# Your Privacy is Our Priority

We believe that data-driven communities can only thrive on a foundation of absolute trust. That's why we've built a "Zero-Knowledge" infrastructure.



**Identity Masking**

Real identities are swapped for randomized cryptographic tokens.



**Data Aggregation**

Insights are only generated from aggregated cohorts, never



STEP 1 OF 3

33% Complete

## Select Data Type

Choose the technical category that best categorizes your data entry.



**Environment**

Air quality, water levels, and biodiversity metrics.



**Infrastructure**



Community Data



## Local Insights

Real-time community health and resource monitoring.

ENVIRONMENTAL IMPACT

-12.4% YoY

### Carbon Footprint Index



 Profile



**Alex Thompson**  
Senior Technical Lead

 **Verified Member**

ACCOUNT CONFIGURATION

 **Account Security** 

Password, 2FA, and authentication methods



**Data Submitted  
Successfully**

Your contribution has been  
anonymized and added to the  
community pool.

## Community Data

 ENTERPRISE GRADE ENCRYPTION

### Welcome Back

Secure access to technical analytics and  
community insights.

EMAIL ADDRESS



name@company.com

PASSWORD

[Forgot Password?](#)



.....



Login →

## 7. Conclusion

The Community Data System demonstrates how a modest, well-architected web application can meaningfully advance SDG 11 by making community data collection and analysis safe, simple, and actionable. Through an Agile SDLC, a clear three-tier architecture, a privacy-first prototype, disciplined GitHub collaboration, and a realistic project plan, Group 3 has produced a fully documented digital public good ready for prototype demonstration and further development beyond this module.

## References

United Nations. (2015). Transforming our world: the 2030 Agenda for Sustainable Development. Sustainable Development Goal 11 – Sustainable Cities and Communities.

Limkokwing University of Creative Technology, Sierra Leone. (2026). PROG 102 – Principles of Software Engineering: Final Project Brief.

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